Why Cross-Dataset Flow-Based Intrusion Detection Fails: An Eight-Dataset Benchmark and Negative Results for Domain-Invariant Transfer

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024

Abstract

Machine-learning detectors of malicious network flows reach 0.985 ROC-AUC when trained and tested on the same dataset and then lose most of that skill on another network. We build a benchmark of eight public flow datasets from five laboratories and three flow extractors, harmonised to features of the same physical meaning, and evaluate seventeen methods and variants in 42 ordered source–target transfers and seven leave-one-dataset-out (LODO) tasks over three seeds. Classical tree ensembles score 0.54 to 0.56 ROC-AUC on average across transfers, 36 to 48% of the transfers are below chance, and even transfer between two datasets from the same laboratory is below chance (0.44 for CatBoost). We designed a domain-invariant method, RIFT, with per-domain rank normalisation anchored on benign traffic, invariance-driven feature selection and test-time label-shift correction. It does not work: it is below its own gradient-boosting baseline (0.484 against 0.557 pairwise ROC-AUC, 0.519 against 0.630 LODO), and every ablation points to the transductive components, in particular its estimate of the target attack prior, which has a mean absolute error of 0.41. CORAL and DANN do not improve on plain gradient boosting either. Controls rule out the most obvious implementation explanations: the same pipeline reaches 0.985 in-domain, no common flow feature keeps the sign of its association with the attack label across the datasets (0 of 14), a classifier recognises the source dataset of a flow with 91.5% accuracy, and nearest-neighbour agreement in the region where datasets overlap is near chance (median Cohen's κ 0.03 over 19 pairs). Source pre-training does not help the few-label setting on average: fine-tuning a source-trained model on 0.1% of the target labels reaches 0.835 ROC-AUC, below the 0.928 of a model trained on those labels alone. We conclude that these datasets disagree on the labelling function of flows and not only on feature scales, and that methods that assume a fixed $P(y \mid x)$ should not be expected to repair it. The package, configurations and all raw results are public.

1 Introduction

Flow-based network intrusion detection is usually evaluated by splitting one dataset into a training and a test part. Under that protocol gradient-boosted trees exceed 0.99 ROC-AUC on CIC-IDS2017 and USB-IDS-1. A growing literature shows that the same models degrade sharply on a second dataset [2, 5, 8, 9, 15], and even datasets from one laboratory transfer poorly. Two explanations are common. The first is covariate shift: capture tools, link speeds and traffic mixes change the scale and distribution of features, which normalisation or domain adaptation could repair [12, 16, 22]. The second is that the datasets define attacks differently, so that the same flow statistics carry different labels, which no unlabelled adaptation can repair.

This paper separates the two as far as a benchmark allows, and it reports a negative result in full. We make four contributions.

1. **A benchmark** of eight public datasets (Table 1) with a common flow schema, a bottom-k hash sampler that gives a reproducible uniform sample of distinct rows, and a protocol with in-domain, pairwise (42 ordered pairs, labelled by whether the pair shares a laboratory or an extractor) and LODO settings, in inductive and transductive modes, over three seeds.

2. **A method designed to work, and its failure.** RIFT combines per-domain rank normalisation anchored on benign traffic, invariance-driven feature selection, and test-time label-shift correction. We evaluate it with a full ablation and show that it is below its own baseline.

3. **Controls** that answer the objection that a negative result may be an implementation failure: positive controls, an oracle search over model family, normalisation and feature subsets, a nearest-neighbour test of whether the same flow statistics carry the same labels across datasets, a dataset-identity classifier, and a fair few-label comparison.

4. **An honest account of what the controls do and do not show**, including that the oracle search covers only two pairs and that the concept-shift evidence is correlational.

A companion study builds a few-label meta-learner on the diagnosis made here; this paper is self-contained.

2 Related work

Layeghy and Portmann [15] train on one NetFlow-v2 dataset and test on another and report attack-class F1 below 1% for several pairs of CIC-IDS2018, UNSW-NB15 and ToN-IoT, with strongly asymmetric results; a domain-invariance method of the same group [16] improves some pairs. Cantone et al. [5] report MCC between 0.26 and 0.44 when CIC-IDS2017 and CIC-IDS2018 are exchanged, even though both come from the same laboratory. D'hooge et al. [8, 9] analyse inter-dataset generalisation per attack class and the effect of data-damaged models on it. Catillo et al. [7] study transfer from CIC-IDS2017 to USB-IDS-1 [6], which share an extractor, and Apruzzese et al. [2] cross-evaluate per-attack detectors across several networks. Engelen et al. [11] document labelling errors in CIC-IDS2017, and Apruzzese et al. [3], Arp et al. [4] discuss pitfalls of machine-learning evaluation in security. Sarhan et al. [20] propose standard NetFlow feature sets to ease cross-dataset use. Our benchmark differs in covering eight datasets and three extractors with one schema, in labelling each pair by how much infrastructure the two datasets share, in running label-free transfer methods against tuned baselines over three seeds, and in reporting the controls that decide whether the negative result is a finding. On the method side, we compare against domain-adversarial training [12] and correlation alignment [22], and RIFT uses the expectation-maximisation correction of the output prior of Saerens et al. [19].

3 Benchmark

Datasets. Table 1 lists the datasets: CIC-IDS2017 [21] and CIC-IDS2018, USB-IDS-1 [6], InSDN [10], UNSW-NB15 [17], ToN-IoT [1], IoT-23 [14] and CTU-13 [13]. They come from five laboratories (UNB, Sannio, UCD, UNSW Canberra and the Stratosphere laboratory of CTU) and three extractors (CICFlowMeter, Argus and Zeek). CIC-IDS2017 and CIC-IDS2018 share a laboratory and an extractor, UNSW-NB15 and ToN-IoT share a laboratory, and IoT-23 and CTU-13 share a laboratory. The attack prior of the distinct rows ranges from 0.15 (CIC-IDS2017, CIC-IDS2018) to 0.93 (USB-IDS-1). Labels are binary, because attack taxonomies differ across datasets; the raw attack type is

Table 1: Datasets. The attack prior is that of the distinct rows after harmonisation. CTU-13 supports only the minimal feature tier and is excluded from the seven-dataset benchmark.

Dataset	Laboratory	Extractor	Attack prior	Feature tier
CIC-IDS2017	UNB CIC	CICFlowMeter	0.152	core
CIC-IDS2018	UNB CIC	CICFlowMeter	0.149	core
USB-IDS-1	Univ. of Sannio	CICFlowMeter	0.930	core
InSDN	UCD	CICFlowMeter	0.640	core
UNSW-NB15	UNSW Canberra	Argus, Bro	0.464	core
ToN-IoT	UNSW Canberra	Zeek	0.743	core
IoT-23	Stratosphere, CTU	Zeek	0.486	core
CTU-13	Stratosphere, CTU	Argus	0.570	minimal

kept for per-family analysis. CTU-13 supports only a minimal feature tier and is therefore used only in the companion study, so the benchmark proper has seven datasets and 42 ordered pairs.

Common flow schema. Extractors export different features, so only features with the same physical meaning and units are used. The core tier has five base features (duration, forward and backward packets, forward and backward bytes) and nine derived rates and ratios, all computed after harmonisation and passed through $\log(1 + x)$. Ports and protocol are excluded because two of the datasets do not export them. Window-size and inter-arrival features exist only for some extractors and are excluded from the core tier. Alias tables resolve the different spellings of CICFlowMeter columns, repeated header rows of CIC-IDS2018 are removed, and negative sentinel values are clipped to zero.

Sampling. Each cleaned row receives a seeded 64-bit hash of its rounded base features and label. Per label we keep the 100 000-row budget of rows with the smallest hash (a bottom-k sketch), so that duplicated flows occupy one slot and the sample is uniform over *distinct* rows. The attack prior is defined on distinct rows, and a class that is short shrinks the sample instead of being back-filled. This makes the benchmark a distinct-row benchmark, so absolute numbers are not comparable with studies that keep duplicates.

Protocol. Each dataset is split 70/30, stratified and seeded. Sources train on their training split and the target is evaluated on its test split. In the transductive mode a method also sees the unlabelled features of the target training split, disjoint from the evaluated rows; in the inductive mode it sees nothing from the target. The in-domain setting is the upper reference. The pairwise setting uses all 42 ordered pairs of the seven core datasets, each labelled *same family* (same laboratory, 4 pairs), *same extractor* (different laboratory, 12 pairs) or *cross-extractor* (26 pairs). In LODO a method trains on the other six datasets pooled. Target labels are never used for training, model selection or threshold tuning; threshold-dependent metrics use the decision rule of the method. We run seeds 0, 1 and 2 and average seeds within a scenario before computing statistics over scenarios. Metrics are ROC-AUC (headline), PR-AUC, F1, balanced accuracy, MCC, recall and false-positive rate.

Methods. Baselines use a global standard scaler fitted on the sources: logistic regression, random forest (300 trees), histogram gradient boosting (HGB), CatBoost [18] with 500 iterations, XGBoost,

and a two-layer MLP. We add CORAL [22] followed by HGB, and DANN [12]; both are transductive. RIFT is described next.

4 RIFT: a method designed to work

RIFT targets three sources of failure separately. *Covariate shift* is addressed by domain-wise normalisation: each domain is mapped to Gaussian ranks computed on its own features. On source domains the normaliser is fitted on benign rows only, because benign traffic is the stable background of a network. On the target, where labels are unknown, the anchor set is the $\gamma = 0.8$ fraction of the $(1 - \hat{\pi}_t)$ lowest-scoring rows under the current estimate $\hat{\pi}_t$ of the target attack prior, and the target is transported into the source representation by quantile matching against an equally truncated reference of source benign rows. *Spurious dataset-specific cues* are addressed by invariance-driven feature selection: a feature is kept only if the sign of its univariate association with the label is identical in every source and its minimum strength is at least 0.05; with a single source, the 25% of features with the largest Kolmogorov–Smirnov distance between source and target are dropped. *Label shift* is addressed by the expectation-maximisation correction of the output prior of Saerens et al. [19], applied after isotonic calibration. The classifier is HGB. Ablations remove one component at a time (domain normalisation, anchor, selection, prior correction), replace the rank transform by robust or z-score scaling, start the anchor from the pooled rather than the source normaliser, and add uncertainty-gated self-training.

5 Results

5.1 In-domain skill does not transfer

Table 2 gives ROC-AUC by setting. In-domain, the tree ensembles reach 0.985 averaged over the seven datasets (CatBoost between 0.950 on IoT-23 and 1.000 on ToN-IoT and USB-IDS-1), logistic regression 0.923. Across the 42 pairs the same methods reach 0.54 to 0.57: CatBoost 0.559, HGB 0.557, XGBoost 0.550, DANN 0.565, random forest 0.538. Between 36% (HGB) and 48% (random forest) of the pairs are below 0.5, which means that the learned rule ranks attacks as *less* suspicious than benign flows on the target. Only 3 of 42 pairs reach 0.9 for CatBoost, and all three involve USB-IDS-1 (the pairs with ToN-IoT, InSDN and USB-IDS-1); the extreme pair is CIC-IDS2018→CIC-IDS2017 at 0.228. Pairs from the same laboratory, which are the easiest on paper, are the worst: CatBoost scores 0.44 on the four same-family pairs, 0.64 on same-extractor pairs and 0.54 on cross-extractor pairs. LODO is better because six sources are pooled, but still 0.65 for CatBoost, and it varies from 0.968 on USB-IDS-1 to 0.333 on IoT-23 (Table 3).

No method is consistently best. A Friedman test over the 42 pairs finds a difference among the 17 transductive variants ($p = 0.006$), but that is driven by the weak methods (logistic regression and the RIFT variants); among the inductive methods ($p = 0.74$) and in LODO ($p = 0.065$ transductive, 0.47 inductive) it finds none. The differences among CatBoost, HGB, XGBoost and DANN are of the order of 0.01, far below the spread between pairs. CORAL and DANN do not repair transfer: pairwise they are at 0.524 and 0.565, in LODO at 0.631 and 0.613, against 0.630 for HGB.

5.2 RIFT fails

Table 4 shows RIFT and its ablations. RIFT is below its own HGB baseline in both settings (pairwise 0.484 against 0.557, LODO 0.519 against 0.630). Removing domain normalisation recovers the baseline (0.558 pairwise, 0.620 LODO), and removing the benign anchor recovers part of it

Table 2: ROC-AUC, mean over scenarios (seeds averaged within a scenario). Pairwise columns split the 42 ordered pairs by what the two datasets share. LODO: train on the other six datasets pooled. Transductive mode; the non-adaptive methods are unaffected by the mode.

Method	In-domain (7)	Same family (4 pairs)	Same extractor (12 pairs)	Cross-extractor (26 pairs)	LODO (7)
LR	0.923	0.341	0.472	0.503	0.480
RF	0.981	0.478	0.597	0.521	0.574
HGB	0.985	0.480	0.643	0.529	0.630
CatBoost	0.985	0.440	0.637	0.541	0.652
XGBoost	0.985	0.397	0.642	0.532	0.612
MLP	0.981	0.453	0.621	0.539	0.503
CORAL	0.985	0.494	0.494	0.542	0.631
DANN	0.980	0.516	0.678	0.521	0.613
RIFT	0.985	0.501	0.465	0.491	0.519
RIFT + self-training	0.985	0.519	0.448	0.486	0.500

Table 3: In-domain and leave-one-dataset-out ROC-AUC per target (inductive).

Target	In-domain	LODO CatBoost	LODO HGB
CIC-IDS2017	0.999	0.803	0.775
CIC-IDS2018	0.970	0.656	0.684
InSDN	0.998	0.672	0.657
IoT-23	0.950	0.333	0.308
ToN-IoT	1.000	0.667	0.560
UNSW-NB15	0.978	0.468	0.467
USB-IDS-1	1.000	0.968	0.957

(0.538, 0.543), so the transductive normalisation is what hurts. Replacing the rank transform by robust or z-score scaling does not help (0.479 and 0.485 pairwise). Self-training does not help (0.478). In the inductive mode, where there is no target pool and the transductive components are inactive, RIFT reduces to feature selection with HGB and all its ablations give the same numbers; it is also below HGB in LODO (0.586 against 0.630).

The label-shift correction fails because its input fails. Across the 42 pairs RIFT estimates the target attack prior with a mean absolute error of 0.41 (bias −0.35, Pearson correlation with the true prior 0.07); in LODO it estimates an average prior of 0.003 where the true average is 0.51. The anchor is meant to be benign traffic, but its purity (the fraction of truly benign rows, logged as a diagnostic and never used by the method) is 0.50 in pairwise and 0.47 in LODO transfer, so the anchor is as contaminated with attacks as the pool. Estimating the prior with a normaliser fitted on the pool ("pooled init.") does not help (0.487). Because the correction expects pure label shift, a source classifier that is inverted on the target gives the EM nothing to correct.

5.3 Source data do not help in the few-label setting

As a reference outside the label-free benchmark, we continue a source-pretrained CatBoost on a labelled budget of 0.1%, 1% and 5% of the target training split (on average 62, 617 and 3 083 rows), and compare it with CatBoost trained on those rows alone. At 0.1%, the source-pretrained model reaches 0.835 and the target-only model 0.928 (Table 5). Giving the continued model 300 or 500 extra trees narrows the gap to 0.911 and 0.914 against 0.932, so the effect is not only a capacity

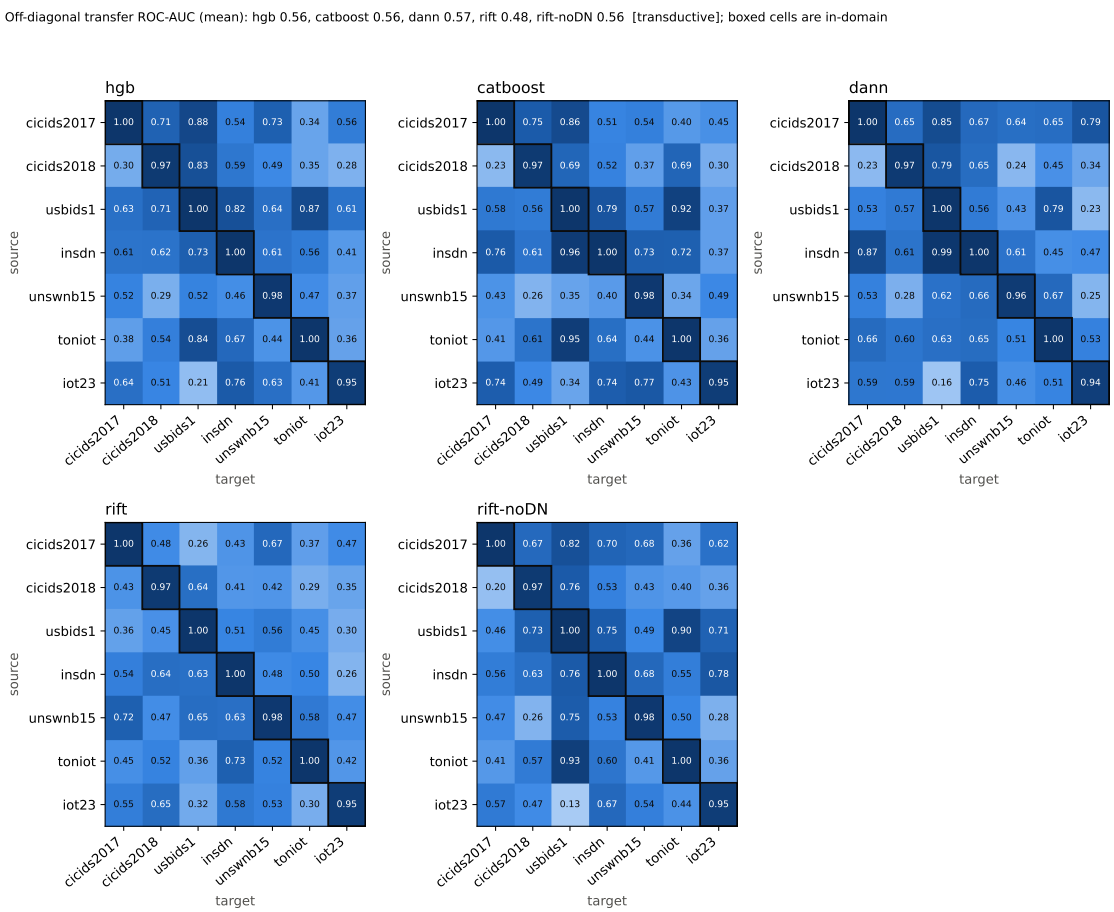


Figure 1: Source→target ROC-AUC of the benchmark methods (transductive mode); the boxed diagonal is the in-domain value.

artefact, but it does not reverse: at 1% and 5% the two are within 0.001 to 0.002 once the trees are matched. Source pre-training thus gives no average benefit over target labels, and a harm at the smallest budget (0.09 AUC with 100 trees, 0.02 with 500).

5.4 Shift and family explanations

An unsupervised shift score (an isolation forest fitted on the source and applied to the target pool) does not predict transfer skill: the Spearman correlation with zero-shot ROC-AUC over 2 142 (scenario, method, seed) points is -0.055. Neither does sharing attack types: for every family we compared the informedness of a method on a target family when the source contains that family and when it does not. Pooled over families (transductive mode), the paired difference for CatBoost is $+0.012$ (95% half-width 0.088, $p = 0.82$) and for HGB -0.046 ($p = 0.14$); none of the 248 (method, mode, family) tests has $p < 0.05$. The benchmark thus shows no evidence that the failure is due to missing attack types in the source.

6 Controls

A negative result is a finding only if competing explanations are ruled out.

Table 4: RIFT and its ablations, transductive mode. Pairwise: 42 pairs; LODO: 7 targets. Prior error: mean absolute error of the estimated target attack prior.

Method	Pairwise AUC	Pairwise bal. acc.	LODO AUC	Prior error
HGB	0.557	0.533	0.630	–
RIFT	0.484	0.510	0.519	0.413
RIFT − domain norm.	0.558	0.536	0.620	0.385
RIFT − benign anchor	0.538	0.521	0.543	0.381
RIFT − invariant selection	0.483	0.495	0.519	0.429
RIFT − prior correction	0.484	0.509	0.519	–
RIFT, pooled init.	0.487	0.500	0.498	0.421
RIFT, robust scaling	0.479	0.504	0.487	0.425
RIFT, z-score scaling	0.485	0.498	0.510	0.414
RIFT + self-training	0.478	0.507	0.500	0.409

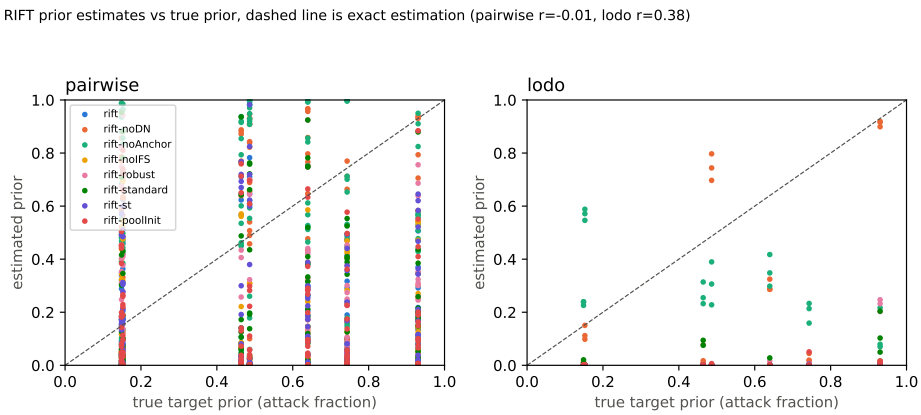


Figure 2: Estimated against true target attack prior for the RIFT variants.

The pipeline works. The same pipeline reaches 0.985 in-domain, and a target-only CatBoost with 0.1% of target labels reaches 0.93, so the features carry the signal and the code can learn it.

Tuning is not the explanation (two pairs). For two pairs we searched over model family (HGB, CatBoost, XGBoost, random forest, logistic regression), main hyperparameters, normalisation (none, global, per-domain rank) and feature subsets (all, or dropping one feature group), and selected the best configuration on the target test split, which is an optimistic oracle. For CIC-IDS2017→CIC-IDS2018 the oracle reaches 0.799 (benchmark HGB 0.683), still far from the in-domain 0.97 to 1.00; for CIC-IDS2017→USB-IDS-1 it reaches 0.862 (benchmark HGB 0.972). Two pairs and a reduced grid are not an exhaustive search, and we do not claim that no configuration exists that does better.

The same statistics do not carry the same labels. We standardise source and target with source statistics, find for each target row its 10 nearest source rows, and restrict to the *overlap region* (rows whose mean neighbour distance is at most the 90th percentile of source-to-source distances). There the majority label of the neighbours should agree with the true target label if $P(y \mid x)$ were shared. Over the 19 pairs with at least 100 overlapping rows the median Cohen's κ is 0.03, 84% of the pairs have $\kappa \leq 0.1$ and the maximum is 0.25 (Fig. 3); the in-domain reference has κ of 0.91 on average. The pair CIC-IDS2017→CIC-IDS2018, from one laboratory, has $\kappa = -0.07$ with 40% of the target rows in the overlap region.

Table 5: Few-label comparison over the 42 pairs, ROC-AUC. The continued model starts from a CatBoost trained on the source and adds the stated number of trees on the labelled target rows; the other model uses only those rows.

Setting	Extra trees	0.1% labels	1% labels	5% labels
Source-pretrained, continued	100	0.835	0.950	0.966
	300	0.911	0.966	0.978
	500	0.914	0.966	0.978
Target labels only	100	0.928	0.964	0.977
	300	0.932	0.967	0.980
	500	0.932	0.967	0.979

No flow feature keeps its meaning. For each of the 14 core features we computed the univariate ROC-AUC with the label in every dataset. The sign of the association (above or below 0.5) is mixed for all 14 features, with per-dataset AUC ranges such as 0.025 to 0.760 for total bytes and 0.057 to 0.808 for forward packets (Fig. 4).

Datasets are easy to tell apart. A CatBoost classifier recognises the dataset of a flow among the seven core datasets with 91.5% accuracy (chance 14.3%), and 91.8% on benign flows alone. Benign traffic therefore also differs between datasets, which is consistent with differences in capture setup and background traffic.

Published results agree. The reported collapses in the literature occur in the same regime: attack-class F1 below 1% for several NetFlow-v2 pairs [15], MCC between 0.26 and 0.44 for CIC-IDS2017 and CIC-IDS2018 [5]. Metrics and feature sets differ across studies, so we compare the pattern and not the numbers.

7 Discussion

The benchmark shows that cross-dataset transfer fails with every method we ran, and that the failure is not repaired by the usual remedies: tuning (on the pairs we tested), normalisation, feature selection, alignment, adversarial invariance, prior correction and self-training. Below-chance ROC-AUC on 36 to 48% of the pairs means that the source rule is not merely less informative on the target but inverted for a large share of the pairs, which a pure covariate shift would not produce. The nearest-neighbour test, the per-feature sign flips and the failure of attack-family overlap to help all point the same way: the datasets disagree on which flows are labelled attacks.

What the evidence does not show. First, the evidence for a change in $P(y \mid x)$ is correlational. Datasets differ in attack mix and benign background, and a per-feature sign flip arises when the attack mix differs, even without a labelling disagreement. The nearest-neighbour test conditions on flow statistics but is restricted to the overlap region, which is small for most pairs (a median overlap fraction below 0.01), and a κ near zero there means only that the neighbours' labels are uninformative, whichever way the shift is decomposed. A per-attack-type comparison of the same attack in two laboratories, which would separate attack-mix shift from labelling disagreement, is not part of this study. Second, the oracle control covers two pairs. Third, the seven datasets are public laboratory captures with their own labelling errors [11], with only 14 harmonised flow features and

no ports or payload, and the distinct-row sampling changes attack priors relative to natural traffic. Fourth, RIFT is one design; the negative result concerns this design and these ablations.

Implications. Evaluation should report cross-dataset results with a pair taxonomy, because same-laboratory pairs can be the worst and one favourable pair hides the rest; a method that is not compared with its own baseline in the same setting, as here, can look useful for the wrong reason. Methods that assume a fixed $P(y \mid x)$ should not be expected to repair these transfers, and the few-label results suggest that a few labelled flows of the target network are worth more than all the source data on average.

8 Conclusion

On eight public datasets, flow classifiers that reach 0.985 ROC-AUC in-domain reach 0.54 to 0.57 across datasets, with about 40% of the transfers below chance, even between datasets from one laboratory. A method designed for the covariate and label shifts of this setting (RIFT) is below its own baseline, and the failure is localised in the transductive components, above all in prior estimation. Controls show that the datasets disagree on the labelling function and not only on scales, though the evidence is correlational and the oracle covers two pairs. The practical consequence is that a handful of labelled flows from the target network is the reliable ingredient.

Reproducibility. The package, configurations, sampled-row hashes and every raw per-run result are in `https://github.com/vinhqdang/cross_ids`. Tables and figures are generated by `make_benchmark_tables.py` and `crossids analyze` from `results/runs.csv` and `results/controls`.

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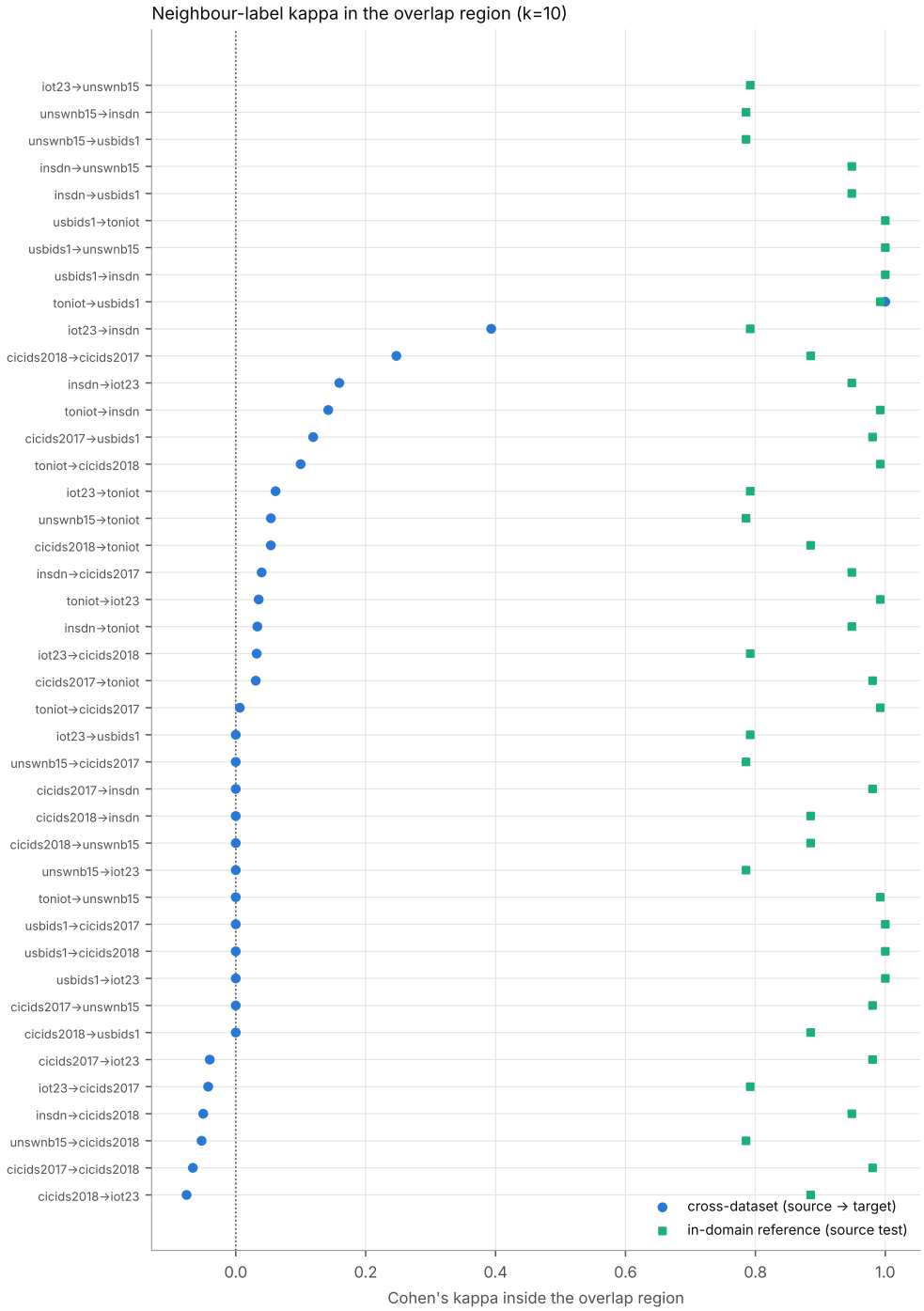


Figure 3: Cohen's κ between the majority label of the nearest source rows and the true target label in the overlap region, per pair, against the in-domain reference.

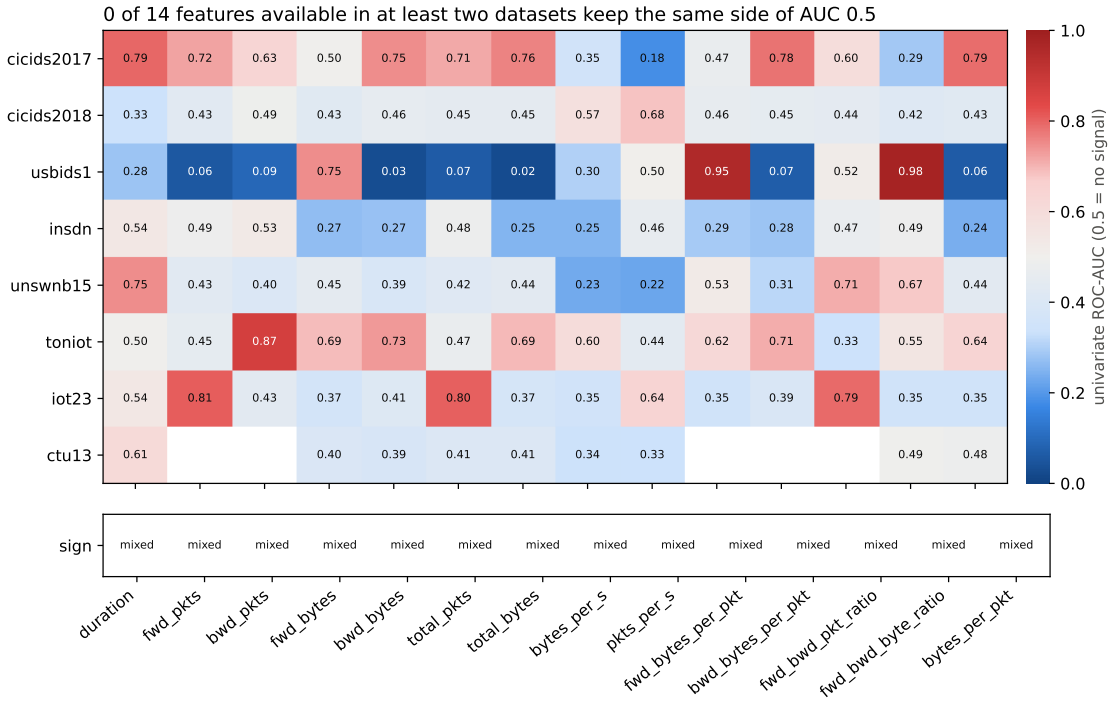


Figure 4: Univariate ROC-AUC of each core feature with the attack label, per dataset (centred at 0.5).